



**Tombstone of
Ludwig Boltzmann
with bust and
entropy formula**

**Zentralfriedhof
(Central Cemetery)
Vienna**

$$\eta = 1 - \frac{Q_2}{Q_1}$$

$$\eta_A (T_1, T_2) \leq \eta_R (T_1, T_2)$$

Chapter 12

The Second Law of Thermodynamics

12.1 The second law

(1) Kelvin-Planck statement (1851):

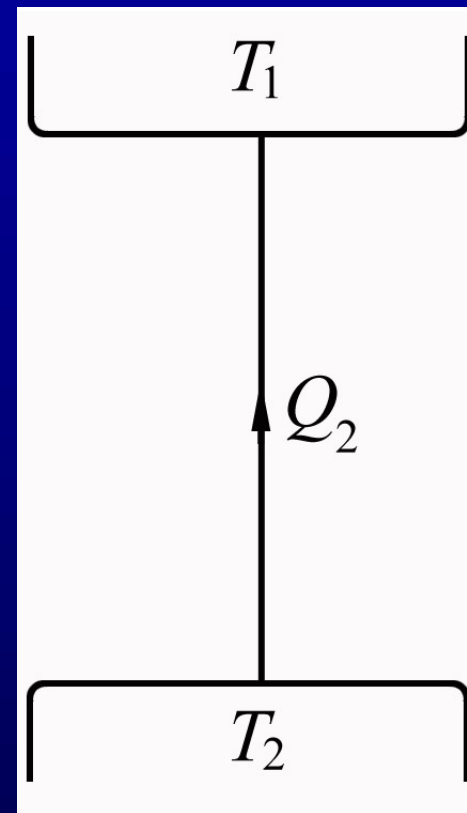
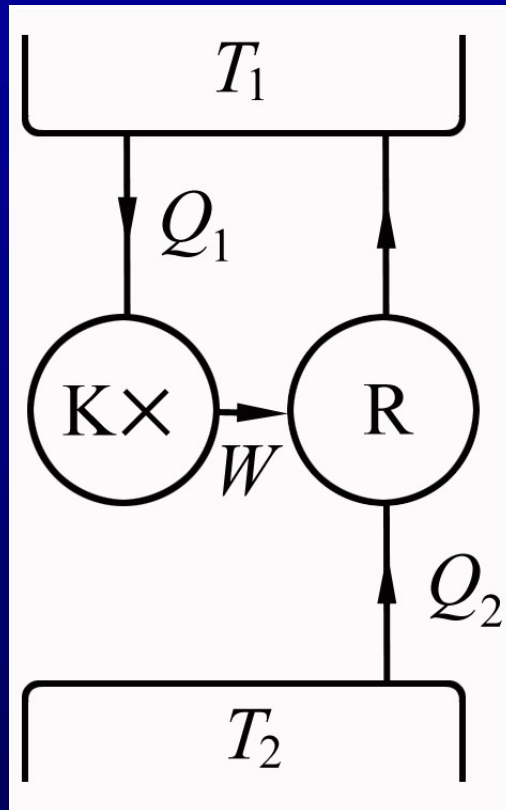
- *No process is possible whose **sole result** is the absorption of heat from a reservoir and the conversion of this heat into work.*

(2) Clausius statement (1850):

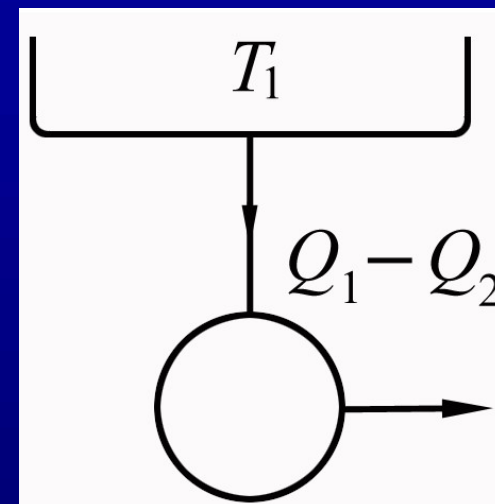
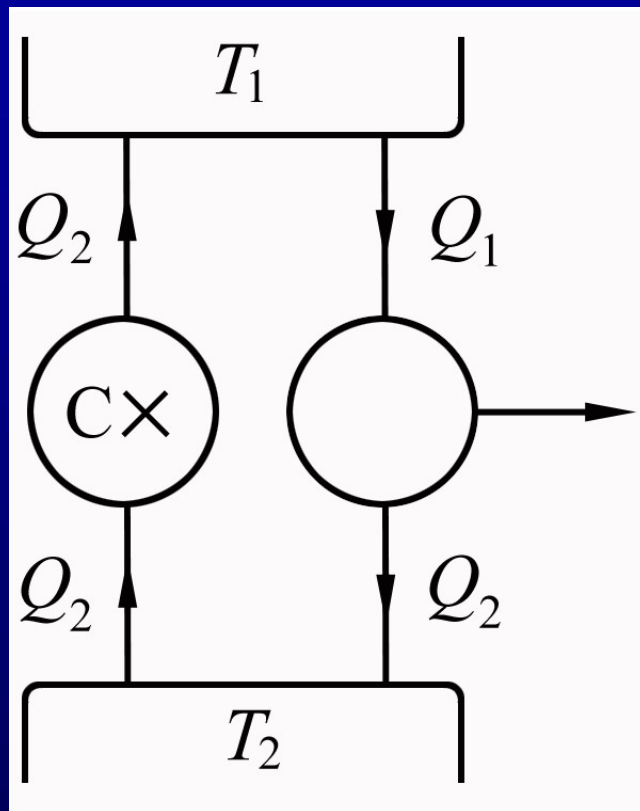
- *No process is possible whose **sole result** is the transfer of heat from a cooler to a hotter body.*

Carathéodory

Equivalence of two statements



$C \times$



K ×

$$C \times \rightarrow K \times$$

$$K \times \rightarrow C \times$$

According to the propositional logic

$$C \equiv K$$

Alternative statement:

- *Perpetual motion machines of the second kind are impossible.*

12.2 Carnot theorem and thermodynamic scale

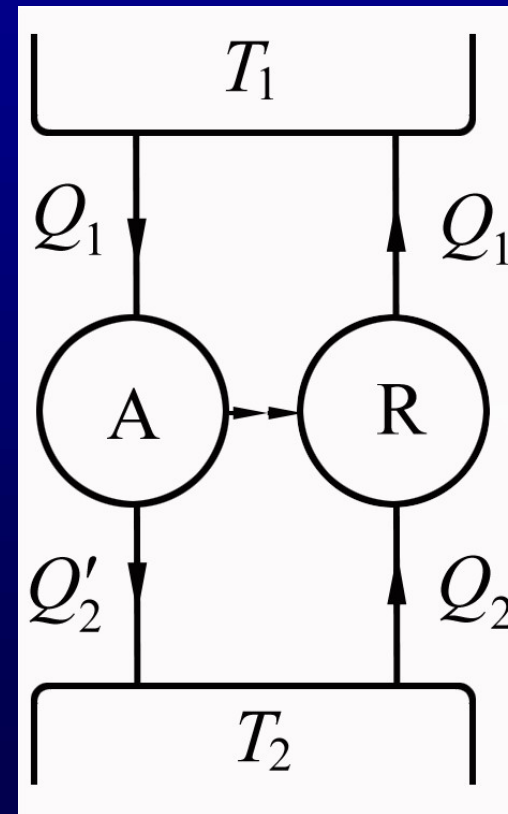
Carnot claimed that $\eta_A (T_1, T_2) \leq \eta_R (T_1, T_2)$

Proof by contradiction

If the theorem is not true,
then $\eta_A > \eta_R$

$$\eta_A = \frac{W'}{Q_1} > \eta_R = \frac{W}{Q_1}$$

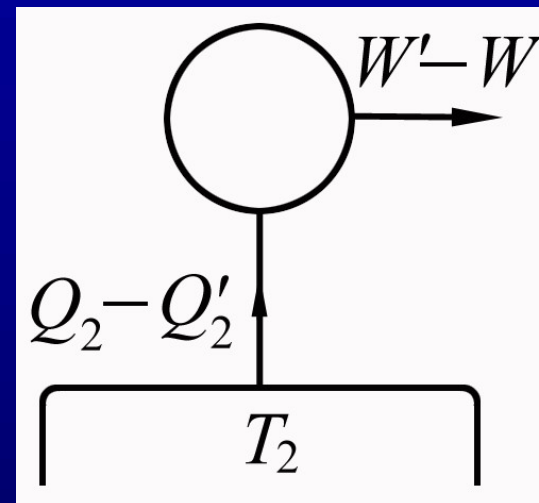
$$\text{i.e., } W' > W$$



$$Q_1 - Q'_2 > Q_1 - Q_2 \quad \text{i.e.,} \quad Q'_2 < Q_2$$

The net effect is

$$Q_2 - Q'_2 \rightarrow W' - W$$



It violates the second law. So the assumption is not true, or

$$\eta_A \leq \eta_R$$

corollary $\eta_{R'}(T_1, T_2) = \eta_R(T_1, T_2)$

1 $T_1, T_2 \rightarrow \eta$ or $\eta \rightarrow T_1, T_2$ **T scale**

2 **IR, R** **definitely different** $\eta_{IR} < \eta_R$

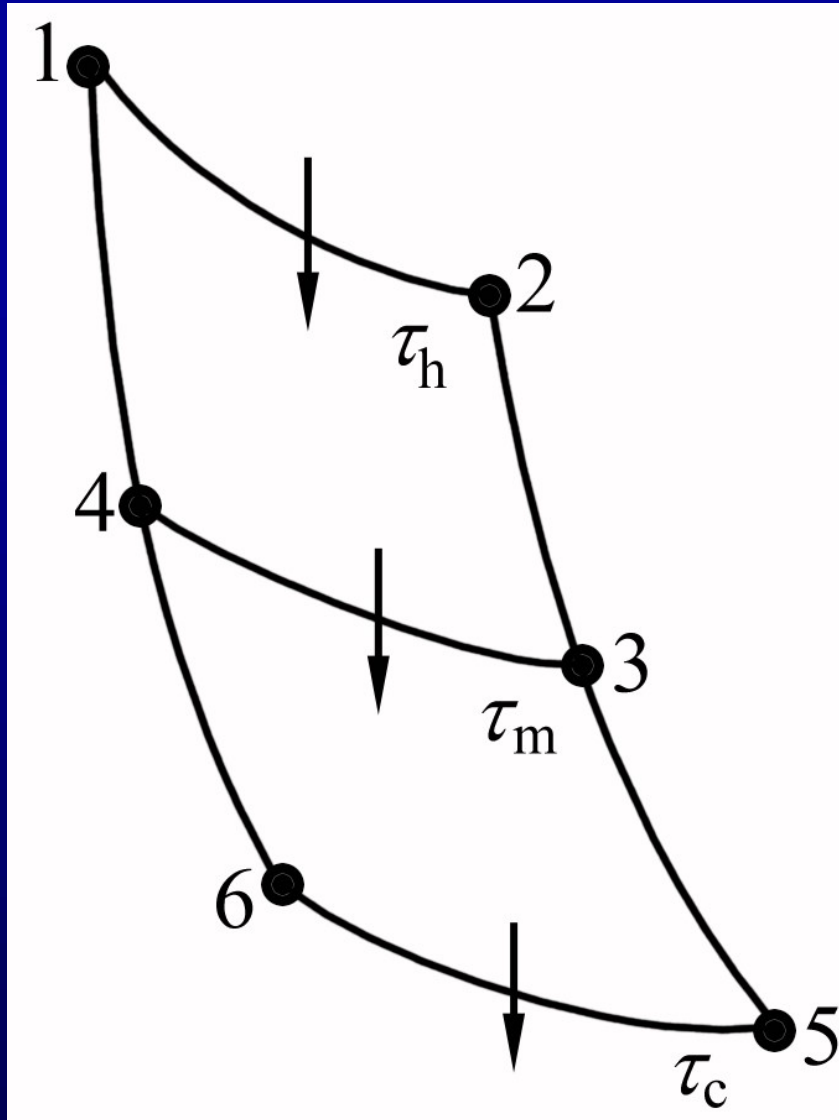
Temperature scale

$$\eta_R = \eta_R(\tau_1, \tau_2)$$

$$\frac{Q_2}{Q_1} = f(\tau_1, \tau_2)$$

$$\eta = 1 - \frac{Q_2}{Q_1}$$

- where τ s may not be calibrated
- working material not mentioned



$$\frac{\Delta Q_{43}}{\Delta Q_{12}} = f(\tau_h, \tau_m)$$

$$\frac{\Delta Q_{65}}{\Delta Q_{43}} = f(\tau_m, \tau_c)$$

$$\frac{\Delta Q_{65}}{\Delta Q_{12}} = f(\tau_h, \tau_c)$$

$$f(\tau_h, \tau_m) f(\tau_m, \tau_c) = f(\tau_h, \tau_c)$$

$$f(\tau_h, \tau_c) = \frac{g(\tau_c)}{g(\tau_h)}$$

Kelvin made a simplest choice $g(\tau) = \tau$

$$\frac{Q_2}{Q_1} = \frac{\tau_2}{\tau_1}$$

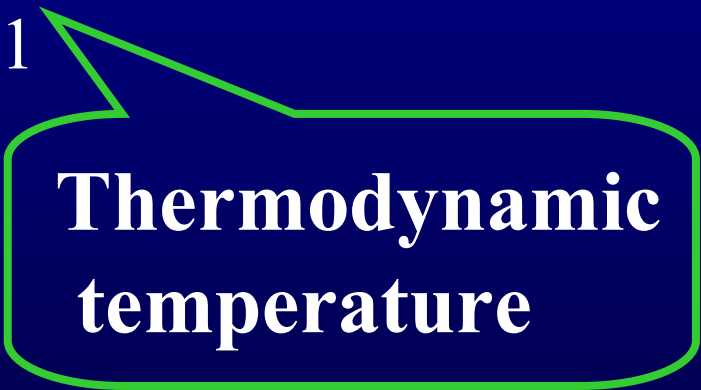
$$\tau_1 = \tau_2 \frac{Q_1}{Q_2}$$

**material-
independent
proportional to
ideal gas scale**

$$\tau = 273.16\text{K} \frac{Q}{Q_{\text{tr}}}$$

$$\frac{\tau}{\tau_{\text{tr}}} = \frac{Q}{Q_{\text{tr}}} \equiv \frac{T}{T_{\text{tr}}}$$

$$\eta \leq 1 - \frac{T_2}{T_1}$$



**Thermodynamic
temperature**

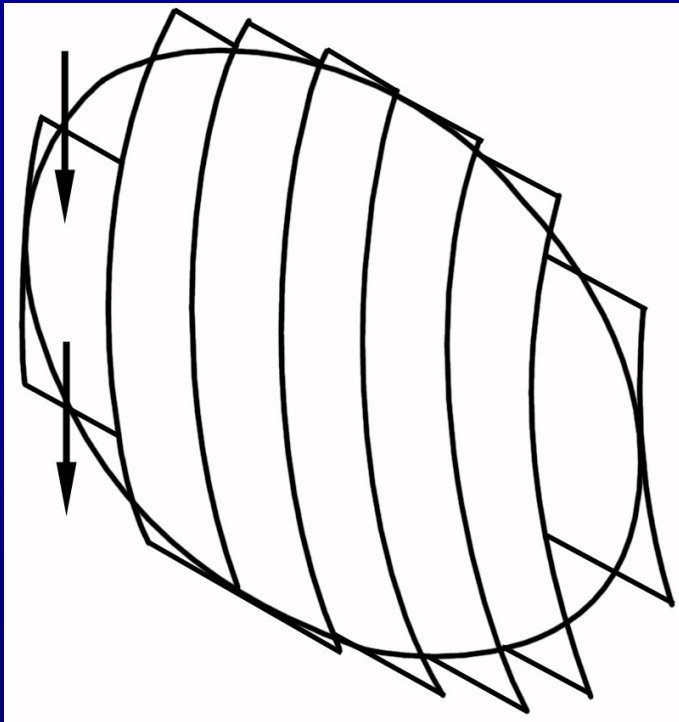
12.3 Entropy and entropy principle

熵

Expelling $Q_2 > 0$  absorbing $Q_2 < 0$

$$\eta_A = 1 + \frac{Q_2}{Q_1} \leq 1 - \frac{T_2}{T_1} = \eta_R$$

$$\frac{Q_1}{T_1} + \frac{Q_2}{T_2} \leq 0 \quad \text{or} \quad \sum_{i=1}^2 \frac{Q_i}{T_i} \leq 0$$



$$\sum_{i=1}^n \frac{Q_i}{T_i} \leq 0 \quad \text{or} \quad \oint \frac{\mathrm{d}Q}{T} \leq 0$$

Clausius inequality

$<$ irreversible; $=$ reversible

For a reversible process

$$\oint \frac{\mathrm{d}Q}{T} = 0$$

exact differential

$$\frac{\mathrm{d}Q}{T} \equiv \mathrm{d}S$$

entropy

$1/T$ is integrating factor of $\mathrm{d}Q$.

$$ydx + xdy = d(xy)$$

$$ydx - xdy = ?$$

$$\frac{1}{y^2} (ydx - xdy) = d(x/y)$$

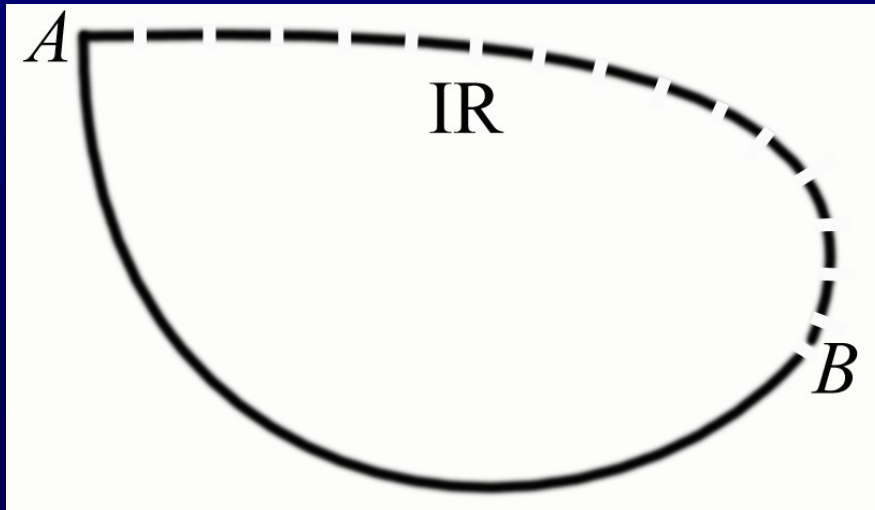
$$\frac{1}{y^2} \text{ is the integrating factor}$$

Exact differential: integral depends on ends (states), process-independent.

Entropy is function of state.

$$S_B - S_A = \int_A^B \left(\frac{\mathrm{d}Q}{T} \right)_R$$

For a cycle involving irreversible process,



$$\oint \frac{\mathrm{d}Q}{T} < 0$$

$$\int_A^B \left(\frac{\bar{d}Q}{T} \right)_{\text{IR}} + \int_B^A \left(\frac{\bar{d}Q}{T} \right)_{\text{R}} < 0$$

$$\int_A^B \left(\frac{\bar{d}Q}{T} \right)_{\text{IR}} - \int_A^B \left(\frac{\bar{d}Q}{T} \right)_{\text{R}} < 0$$

$$S_B - S_A > \int_A^B \left(\frac{\bar{d}Q}{T} \right)_{\text{IR}}$$

For infinitesimal process

$$dS > \left(\frac{\bar{d}Q}{T} \right)_{\text{IR}}$$

Entropy principle reads

$$\boxed{dS \geq \frac{\delta Q}{T}}$$

For an adiabatic process $\delta Q = 0$
reversible $dS = 0$ **isentropic**
irreversible $dS > 0$

For an isolated system (any process) $\delta Q = 0$
reversible $dS = 0$
irreversible $dS > 0$

Isolated system  **Equilibrium state**
 **maximum entropy**

Entropy change or difference

(1) Known the expression for entropy

$$\Delta S = S_f - S_i$$

For example, for an **ideal gas**

$$dS = \frac{1}{T} (dU + p dV) = C_V \frac{dT}{T} + nR \frac{dV}{V}$$

$$S_f - S_i = C_V \ln \frac{T_f}{T_i} + nR \ln \frac{V_f}{V_i}$$

$$S = C_V \ln T + nR \ln V + S_0$$

$$S = C_V \ln T + R \ln V + S_0$$

For free expansion from V to $2V$ ($T = \text{const.}$)

$$\Delta S = nR \ln 2 > 0$$

irreversible

(2) Entropy change for a **reservoir**

$$\Delta S = \int \frac{\mathrm{d}Q}{T} = \frac{1}{T} \int \mathrm{d}Q = \frac{\Delta Q}{T}$$

(3) For two states connected by a quasi-static process

$$\Delta S = \int_{(i)}^{(f)} dS = \int_{(i)}^{(f)} \frac{\bar{d}Q}{T}$$

$$\bar{d}Q \begin{cases} C_P dT \\ C_V dT \\ \dots \end{cases}$$

(4) For two states connected by an irreversible process, *find a reversible process connecting same pair of states.*

For free expansion from T, V_i to T, V_f , image a **reversible isothermal** process connecting same pair of states

$$\Delta S = \int \left(\frac{\delta Q}{T} \right)_{\text{rev}} = \frac{1}{T} \int_{V_i}^{V_f} p dV = \frac{1}{T} \int_{V_i}^{V_f} \frac{nRT}{V} dV = nR \ln \frac{V_f}{V_i}.$$

$dU = 0$

Why the free expansion is irreversible while the isothermal (quasi-static) expansion is reversible ?

For the isothermal expansion, the change of entropy of the system is

$$\Delta S_{\text{system}} = nR \ln \frac{V_f}{V_i}.$$

And the work done by the system is

$$W' = \int_{V_i}^{V_f} p dV = \int_{V_i}^{V_f} \frac{nRT}{V} dV = nRT \ln \frac{V_f}{V_i}.$$

The heat from the reservoir to the system is

$$Q = W' = nRT \ln \frac{V_f}{V_i}.$$

The change of entropy of the reservoir is

$$\Delta S_{\text{reservoir}} = \frac{-Q}{T} = -nR \ln \frac{V_f}{V_i}.$$

The total change in entropy is

$$\Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{reservoir}} = 0.$$

For the free expansion, the change of entropy of the system is

$$\Delta S_{\text{system}} = nR \ln \frac{V_f}{V_i}.$$

The change of entropy of the surrounding is

$$\Delta S_{\text{sur}} = 0.$$

The total change in entropy is

$$\Delta S_{\text{total}} = \Delta S_{\text{system}} + \Delta S_{\text{sur}} = nR \ln \frac{V_f}{V_i} > 0.$$

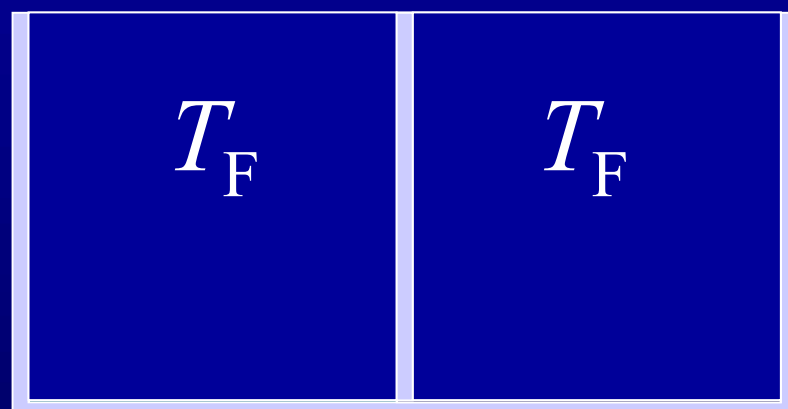
Mixing

$$\Delta Q = m_1 c_1 (T_f - T_1) + m_2 c_2 (T_f - T_2) = 0$$

$$T_f = \frac{m_1 c_1 T_1 + m_2 c_2 T_2}{m_1 c_1 + m_2 c_2}$$

find a **reversible process** connecting same pair of initial (T_1, T_2) and final states (T_f) .

T_1
 $T_1 - \Delta$
 $\cdot$
 T_F



T_F
 $\cdot$
 $T_2 + \Delta$
 T_2

$$\Delta S = \int_{T_1}^{T_f} \frac{m_1 c_1 dT}{T} + \int_{T_2}^{T_f} \frac{m_2 c_2 dT}{T} = \dots$$

Only possible for reversible processes

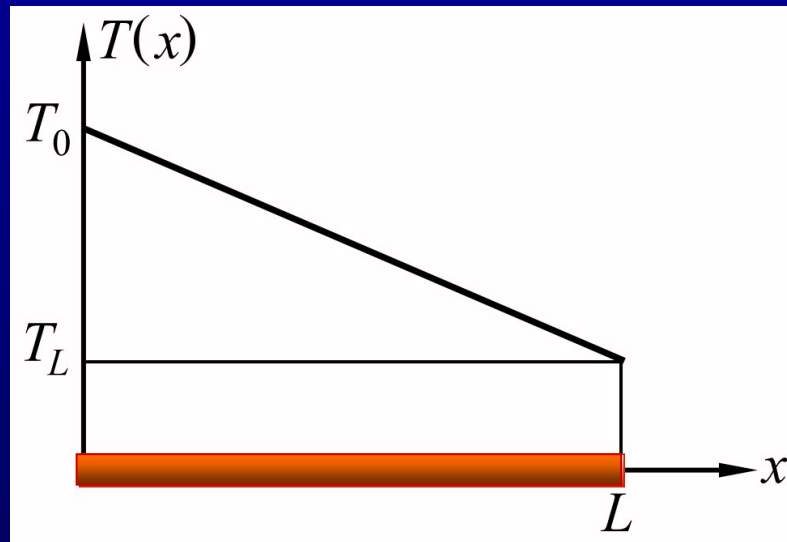
A diffusion of two dissimilar inert ideal gases is equivalent to **two free expansions.**

Diffusion can happen between same gas.

$2 \times nR \ln 2?$ **Gibbs paradox**

(5) From a **non-equilibrium initial state** to an equilibrium final state.

If local entropy can be defined, ...



$$T(x) = T_0 - \frac{T_0 - T_L}{L} x$$

Stationary state: heat flow

When heat baths removed, the bar is in non-equilibrium state and will finally relax to equilibrium state.

Equilibrium temperature can be fixed through

$$\Delta Q = 0$$

$$\Delta Q = \int dq(x) = \int dm c \int_{T(x)}^{T_f} dT$$

$$= c \rho A \int_0^L dx \int_{T(x)}^{T_f} dT = 0$$

$$T_f = \frac{T_0 + T_L}{2}$$

Replacing the relaxation process by a reversible process, the total entropy increase can be calculated as

$$\Delta S = \int_0^L c\rho A dx \int_{T(x)}^{T_f} \frac{dT}{T} \quad T(x) = T_0 - \frac{T_0 - T_L}{L} x$$

$$= \frac{c\rho AL}{T_0 - T_L} \int_{T_L}^{T_0} dT(x) \int_{T(x)}^{T_f} \frac{dT}{T}$$

$$= c\rho AL \left\{ 1 - \frac{1}{T_0 - T_f} \left[T_0 \ln \left(\frac{2T_0}{T_0 + T_L} \right) + T_L \ln \left(\frac{T_0 + T_L}{2T_L} \right) \right] \right\}$$

$$\int_{T_L}^{T_0} \boxed{\text{d}T(x)} \boxed{\int_{T(x)}^{T_f} \frac{\text{d}T}{T}} = T(x) \int_{T(x)}^{T_f} \frac{\text{d}T}{T} \Big|_{T(x)=T_L}^{T_0} - \int_{T_L}^{T_0} T(x) \text{d} \left(\int_{T(x)}^{T_f} \frac{\text{d}T}{T} \right)$$

$\text{d}v$

u

$$= T_0 \int_{T_0}^{T_f} \frac{\text{d}T}{T} - T_L \int_{T_L}^{T_f} \frac{\text{d}T}{T} + \int_{T_L}^{T_0} \cancel{T(x)} \text{d}T(x) \frac{1}{\cancel{T(x)}}$$

$$= T_0 - T_L - \left(T_0 \ln \frac{T_0}{T_f} + T_L \ln \frac{T_f}{T_L} \right)$$

Entropy and irreversibility

Microscopic interpretation of entropy

Boltzmann relation $S = k_B \ln W$

W is number of microscopic states (microstates) corresponding to the macrostate.

In free expansion, before: $W = 1$

after: $W = 2^N$

macrostate: partition removed
microscopic states: every
molecule can itinerate freely

abcd 0(as macrostate 1);

abc d; abd c; acd b; bcd a;

ab cd; bc da; cd ab; ac bd; bd ac; ad bc;

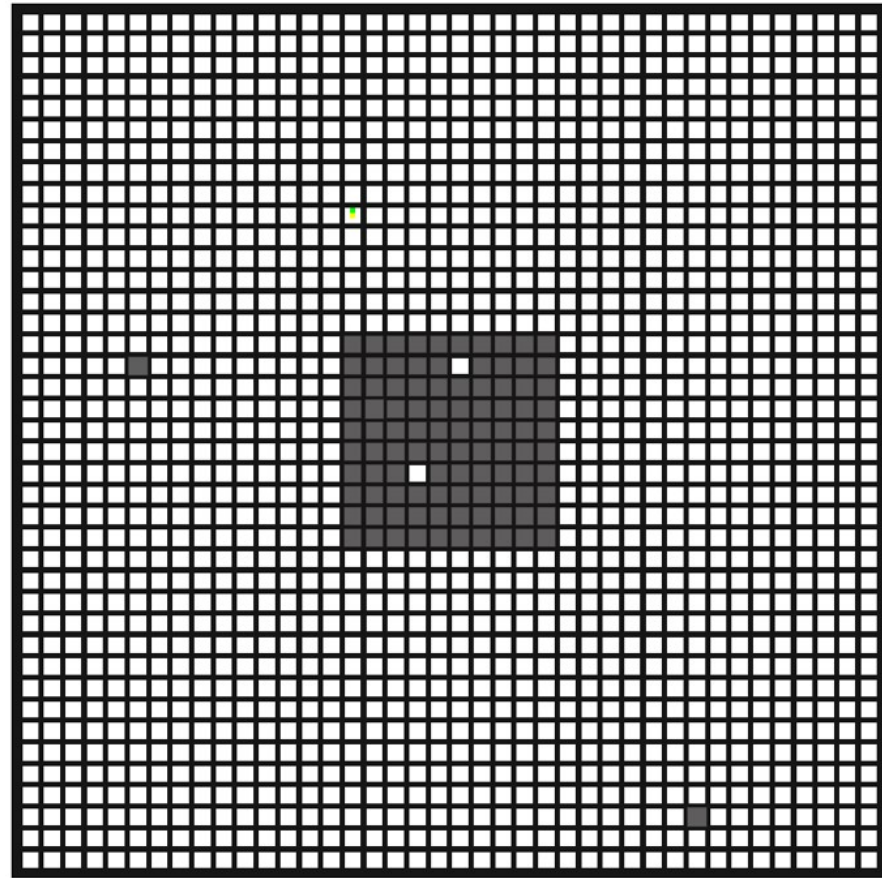
a bcd; b cda; c dab; d abc;

0 abcd.

Problem 12.9

Out 1500

In 100



How many states that correspond to
the macrostate, **out 2 in 98**?

$$C_{100}^2 \times C_{1500}^2$$

Entropy and disorder

$$\frac{\dot{D}(n)}{D(n)} = \frac{\dot{S}(n)}{S(n)} - \frac{\dot{n}}{n \ln n}$$

Time arrow and open system

Entropy and useful energy $F = U - TS$

Heat death of universe

*Entropy and information theory

S. J. Blundell et al., Concepts in Thermal Physics
Chap. 15 p. 157

Assignment : 12.8, 12.9

Zeroth law T

First law U

Second law S

Zeroth law

Everyone knows

First law

3

Second law

2

Third law

1

Fourth law

Nobody can find

Onsanger reciprocal relations

12.4 Thermodynamic Potentials

$$\delta Q = TdS$$

$$dU = TdS - pdV + \mu dn$$

mole number

chemical potential

fundamental equation of thermodynamics

$$\left(\frac{\partial U}{\partial S}\right)_{V,n} = T, \quad \left(\frac{\partial U}{\partial V}\right)_{S,n} = -p, \quad \left(\frac{\partial U}{\partial n}\right)_{S,V} = \mu$$

U is an extensive quantity homogeneous function of 1st degree

$$U(\lambda S, \lambda V, \lambda n) = \lambda U(S, V, n)$$

$$\left. \frac{\partial}{\partial \lambda} \right|_{\lambda=1} \rightarrow \frac{\partial U}{\partial(\lambda S)} S + \frac{\partial U}{\partial(\lambda V)} V + \frac{\partial U}{\partial(\lambda n)} n = U$$

Euler equation

$$U = TS - pV + \mu n$$

$$dU = TdS - pdV + \mu dn$$

**When there exists long-range interaction,
 U ceases to be extensive.**

Euler's Homogeneous Function Theorem

Let $f(x_1, x_2, \dots, x_n)$ be the homogeneous function of degree k so that

$$f(\lambda x_1, \lambda x_2, \dots, \lambda x_n) = \lambda^k f(x_1, x_2, \dots, x_n).$$

Making derivative on both sides of above equation with respect to λ , we have

$$\sum_{i=1}^n \frac{\partial f}{\partial(\lambda x_i)} \frac{\partial(\lambda x_i)}{\partial \lambda} = k \lambda^{k-1} f(x_1, x_2, \dots, x_n).$$

Let $\lambda = 1$, then we have

$$\sum_{i=1}^n x_i \frac{\partial f}{\partial x_i} = k f(x_1, x_2, \dots, x_n).$$

Legendre transformation

$$dU = TdS - pdV - Vdp + Vdp + \mu dn$$

$$d(U + pV) = TdS + Vdp + \mu dn \equiv dH$$

~reaction heat

Legendre transformation

Given the function of two variable x and y

$$\begin{aligned}df &= \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy \\&= u dx + v dy.\end{aligned}$$

To change the differentials dx and dy to du and dy , we make the transformation

$$g = f - ux.$$

Thus, we have

$$\begin{aligned}dg &= df - u dx - x du \\&= \cancel{u dx} + v dy - \cancel{u dx} - x du \\&= -x du + v dy\end{aligned}$$

$$d(U - TS) = -SdT - pdV + \mu dn \equiv dF$$

F is called **Helmholtz free energy**

$$dG = \underline{-SdT + Vdp} + \mu dn$$

~useful work

G **Gibbs free energy**.

$$\mu = \left(\frac{\partial G}{\partial n} \right)_{T,p}$$

$$dU = TdS - pdV + \mu dn$$

$$\mu = \left(\frac{\partial U}{\partial n} \right)_{S,V}$$

$$G(T, p, \lambda n) = \lambda G(T, p, n)$$

$$G = \left(\frac{\partial G}{\partial n} \right)_{T,p} n = \mu n$$

$$dG = -SdT + Vdp + \mu dn = nd\mu + \mu dn$$

$$nd\mu = -SdT + Vdp$$

$$d\mu = -S_m dT + V_m dp$$

**Gibbs-Duheim
equation**

Maxwell relation

$$\begin{aligned}dG &= \left(\frac{\partial G}{\partial T} \right)_{p,n} dT + \left(\frac{\partial G}{\partial p} \right)_{T,n} dp + \left(\frac{\partial G}{\partial n} \right)_{T,p} dn \\&= -SdT + Vdp + \mu dn\end{aligned}$$

From

$$\frac{\partial}{\partial p} \left(\frac{\partial G}{\partial T} \right) = \frac{\partial}{\partial T} \left(\frac{\partial G}{\partial p} \right) = \frac{\partial^2 G}{\partial T \partial p}$$

We have

$$-\left(\frac{\partial S}{\partial p} \right)_{T,n} = \left(\frac{\partial V}{\partial T} \right)_{p,n} - \left(\frac{\partial \mathcal{K}}{\partial m} \right)_{\bar{p}, \bar{p}} \equiv$$

$$dU = TdS - pdV + \mu dn$$

$$\left(\frac{\partial T}{\partial n}\right)_{S,V} = ? - \left(\frac{\partial p}{\partial S}\right)_{V,n} = ? \quad \left(\frac{\partial \mu}{\partial V}\right)_{S,n} = ?$$

$$dU = TdS - pdV$$

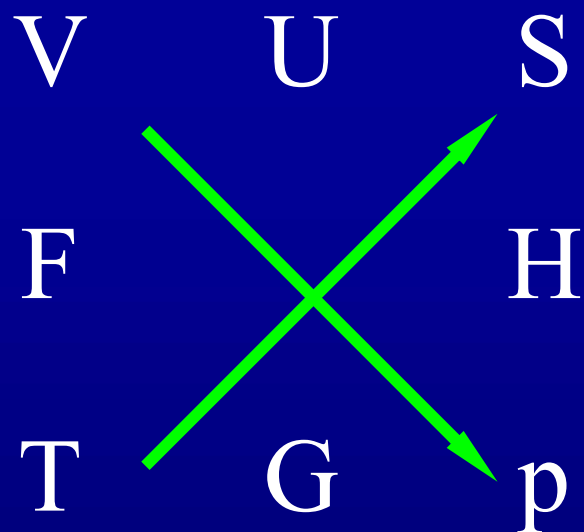
$$dF = -SdT - pdV$$

$$= T \left[\left(\frac{\partial S}{\partial T} \right)_V dT + \left(\frac{\partial S}{\partial V} \right)_T dV \right] - pdV$$

$$= C_V dT + T \left(\frac{\partial p}{\partial T} \right)_V dV - pdV$$

dU is exact differential,

$$\left(\frac{\partial U}{\partial V} \right)_T = T \left(\frac{\partial p}{\partial T} \right)_V - p$$



$$dU = TdS - pdV$$

$$dF = -SdT - pdV$$

$$dH = TdS + Vdp$$

$$dG = -SdT + Vdp$$

Thermodynamic square

**The Friendly Vagabond Usually Shares
His precious Gifts.**

Just for fun. Not recommended

***For a magnetic system**

$$dU = TdS - pdV + \tilde{H}dM$$

$$dG = -SdT + Vdp - Md\tilde{H}$$

One of Maxwell's relations is

piezomagnetic

$$\left(\frac{\partial V}{\partial \tilde{H}}\right)_{T,p} = -\left(\frac{\partial M}{\partial p}\right)_{T,\tilde{H}}$$

magnetostriction

Another Maxwell relation is

$$\left(\frac{\partial S}{\partial \tilde{H}}\right)_{p,T} = \left(\frac{\partial M}{\partial T}\right)_{p,\tilde{H}}$$

$$\text{lhs} \left(\frac{\partial S}{\partial \tilde{H}} \right)_T = - \left(\frac{\partial S}{\partial T} \right)_{\tilde{H}} \left(\frac{\partial T}{\partial \tilde{H}} \right)_S = - \frac{C_{\tilde{H}}}{T} \left(\frac{\partial T}{\partial \tilde{H}} \right)_S$$

Material is supposed to obey Curie's law:

$$M = \frac{CV}{T} \tilde{H}$$

$$\text{rhs} \left(\frac{\partial M}{\partial T} \right)_{\tilde{H}} = - \frac{1}{T} M$$

$$\text{Finally} \left(\frac{\partial T}{\partial \tilde{H}} \right)_S = \frac{M}{C_{\tilde{H}}} > 0$$

Adiabatic demagnetization  $\Delta T < 0$

Cyclic rule (Euler's chain relation)

$$f(x, y, z) = 0 \implies \left(\frac{\partial x}{\partial y} \right)_z \left(\frac{\partial y}{\partial z} \right)_x \left(\frac{\partial z}{\partial x} \right)_y = -1$$

$$z = z(x, y)$$

$$dz = \left(\frac{\partial z}{\partial x} \right)_y dx + \left(\frac{\partial z}{\partial y} \right)_x dy$$

Let $dz=0$ ($z=\text{const}$)

$$0 = \left(\frac{\partial z}{\partial x} \right)_y dx + \left(\frac{\partial z}{\partial y} \right)_x dy \implies \left(\frac{\partial x}{\partial y} \right)_z = - \frac{\left(\frac{\partial z}{\partial y} \right)_x}{\left(\frac{\partial z}{\partial x} \right)_y}$$

Criteria of thermodynamic equilibrium

For an **isolated system** $\Delta S \geq 0$

equilibrium state $S = S_{\max}$

What is the criteria for other systems?

criterion pl. criteria

If any **medium** is involved, near local equilibrium the **total** entropy is

$$\begin{aligned}\Delta S_t &= \Delta S + \Delta S_0 = \Delta S - \frac{\Delta Q}{T_0} \\ &= \Delta S - \frac{\Delta U - \Delta W - \mu \Delta n}{T_0} \geq 0\end{aligned}$$


$$\Delta U \leq T_0 \Delta S + \Delta W + \mu \Delta n$$

$$(\Delta U)_{S, V, n} \leq 0$$

An equilibrium state at fixed (S, V, n) is a state of $U_{\min}$.

Similarly, we have

$$\Delta F = \Delta(U - TS) \leq -S\Delta T - p\Delta V + \mu \Delta n$$

$$\Delta G = \Delta(F + pV) \leq -S\Delta T + V\Delta p + \mu \Delta n$$

$$(\Delta F)_{T, V, n} \leq 0$$

$$(\Delta G)_{T, p, n} \leq 0$$

Maximum work by system

For a system of fixed n

Isothermal ($T = T_0$) reservoir

$$\Delta S_t = \Delta S + \Delta S_{\text{reservoir}}$$

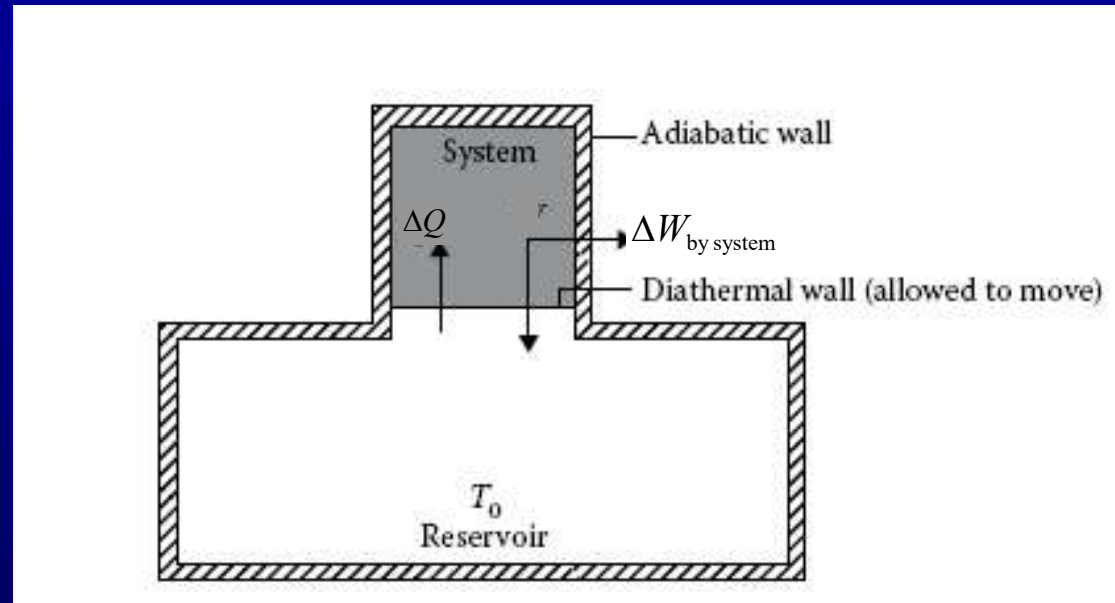
$$= \Delta S - \frac{\Delta Q}{T_0} \geq 0$$

$$\Delta Q = \Delta U - \Delta W$$

$$\Delta W \geq \Delta U - T_0 \Delta S$$

$$\Delta W_{\text{by system}} = -\Delta W \leq -\Delta(U - TS) \equiv -\Delta F$$

The maximum amount of work done by the system in an isothermal process is the decrease of the Helmholtz free energy.



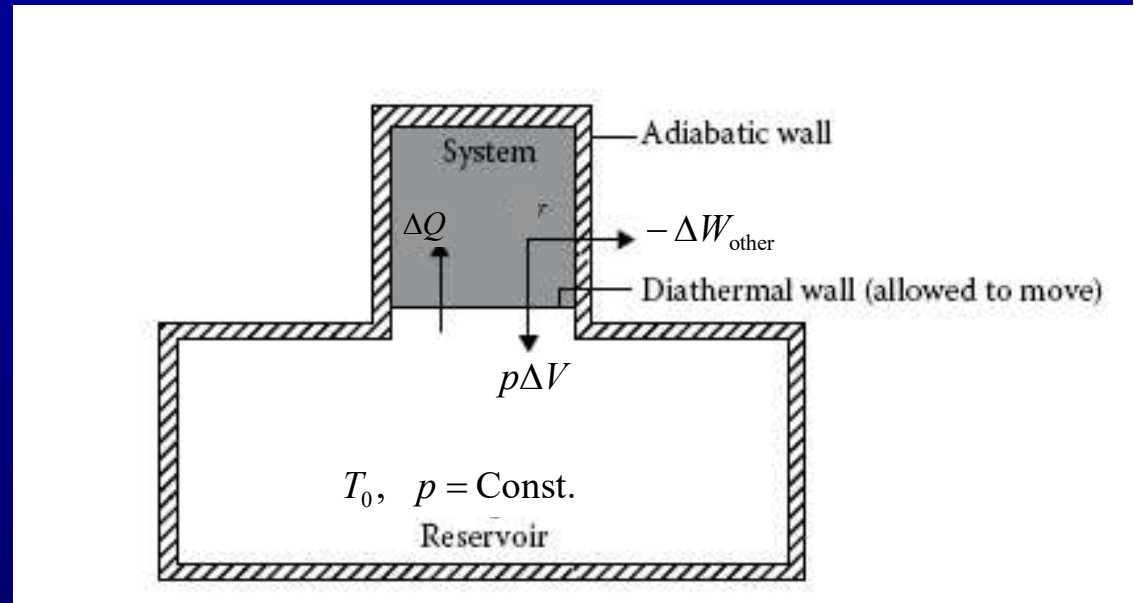
Maximum work by system

isothermal
isobaric

$$\Delta W \geq \Delta U - T_0 \Delta S$$

$$\Delta W = -p\Delta V + \Delta W_{other}$$

non-expansion work

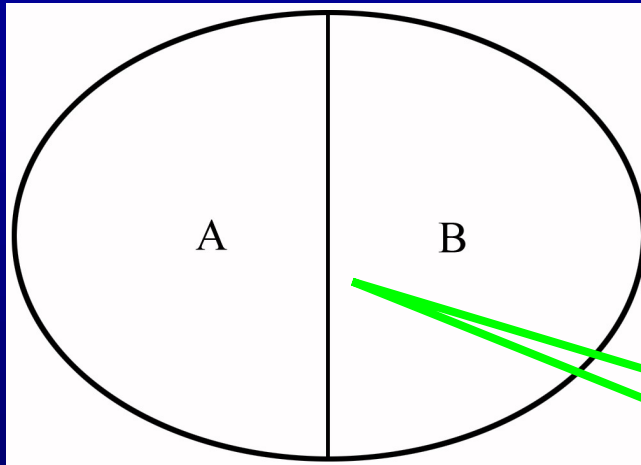


$$-\Delta W_{other} \leq -\Delta(U - TS + pV) \equiv -\Delta G$$

The maximum amount of non-expansion work done by the system in an isothermal and isobaric process is the decrease of the Gibbs free energy.

Local equilibrium

← equilibrium criteria

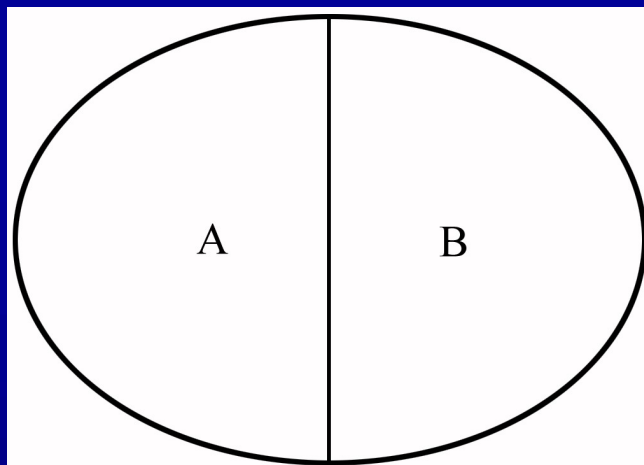


Conditions for equilibrium:

$$T_A = T_B, \quad p_A = p_B$$

$$\mu_A = \mu_B$$

Imaginary or real
phase boundary



A+B isolated
equilibrium $\rightarrow$ maximum
entropy

For a virtual deviation to the equilibrium:

$$\delta U_A, \delta V_A, \delta n_A \quad \delta U_A + \delta U_B = 0, \quad \text{constraints}$$

$$\delta V_A + \delta V_B = 0,$$

$$\delta U_B, \delta V_B, \delta n_B \quad \delta n_A + \delta n_B = 0.$$

$$\delta S = \delta S_A + \delta S_B = 0.$$

condition for
equilibrium

$$dU = TdS - pdV + \mu dn$$

$$dS = \frac{dU}{T} + \frac{p}{T}dV - \frac{\mu}{T}dn$$

$$\delta S_A = \frac{1}{T_A} \delta U_A + \frac{p_A}{T_A} \delta V_A - \frac{\mu_A}{T_A} \delta n_A,$$

$$\delta S_B = \frac{1}{T_B} \delta U_B + \frac{p_B}{T_B} \delta V_B - \frac{\mu_B}{T_B} \delta n_B.$$

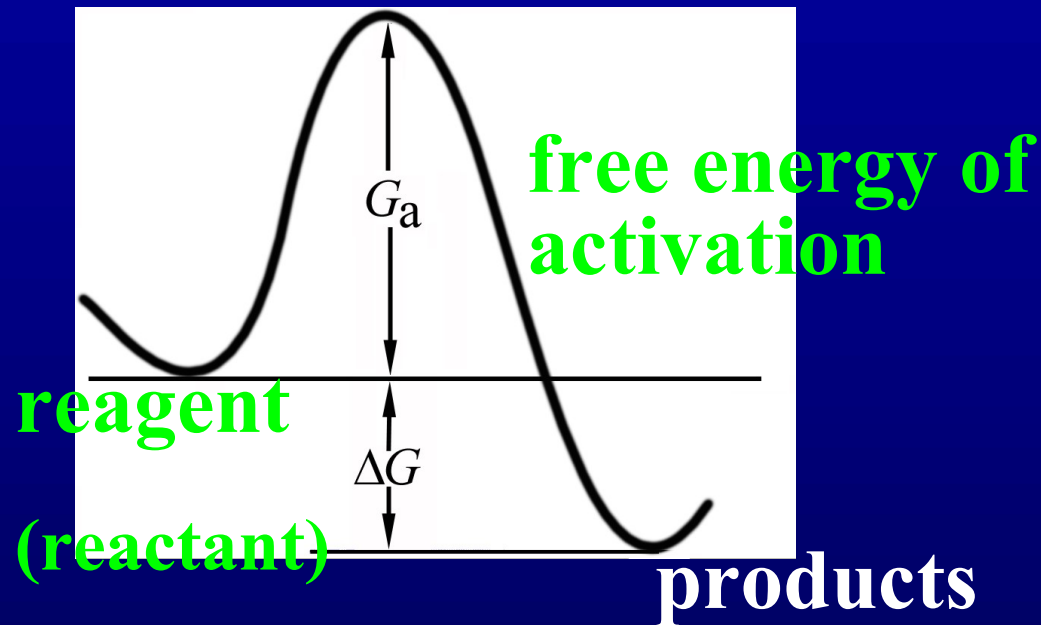
$$\delta S = \left(\frac{1}{T_A} - \frac{1}{T_B} \right) \delta U_A + \left(\frac{p_A}{T_A} - \frac{p_B}{T_B} \right) \delta V_A$$

$$- \left(\frac{\mu_A}{T_A} - \frac{\mu_B}{T_B} \right) \delta n_A = 0.$$

$\delta U_A, \delta V_A, \delta n_A$
can vary independently

$$T_A = T_B, \quad p_A = p_B, \quad \mu_A = \mu_B.$$

$$(\Delta G)_{T, p, n} \leq 0$$



ΔG means the reaction is possible.
(Catalyst lowers G_a)

In open systems,
catalyst also change ΔG

Assignment : 12.12, 12.13, 12.14

**Zeroth law: C. Carathéodory (1909),
(enters textbook) R H Fowler (1939)**

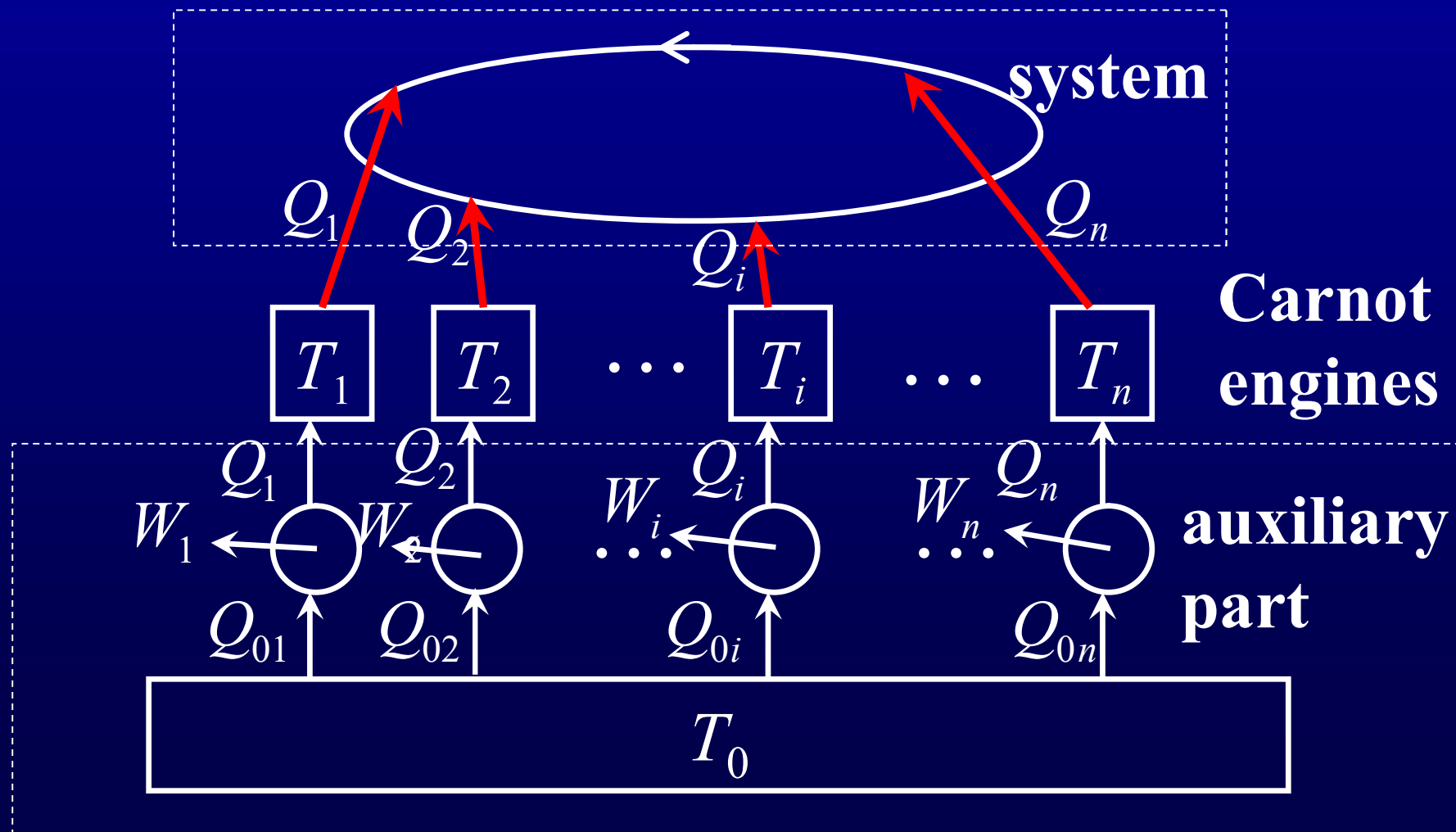
**1st law: J. R. Mayer(1842),
H von Helmholtz(1847),
J. P. Joule(1850)
S. Carnot(1824)**

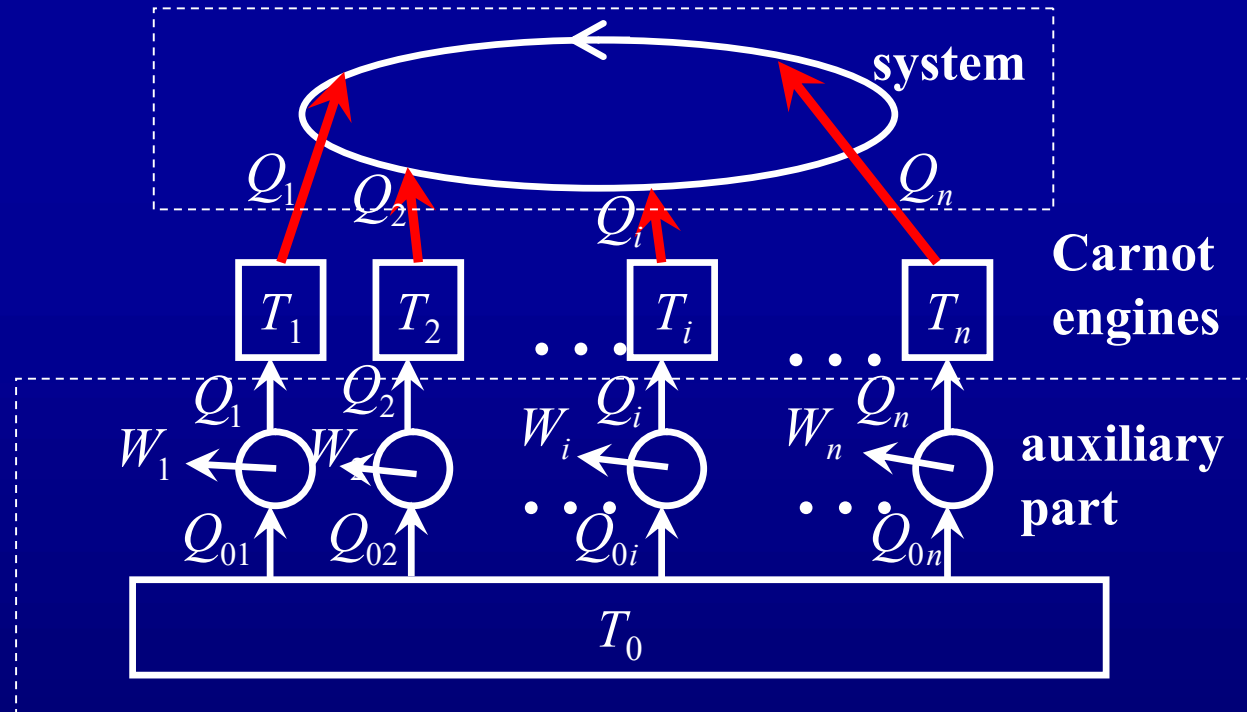
**2nd law: R. J. Clausius(1850),
Kelvin(W. Thomson1851)**

**3rd law: (Nernst theorem 1907)
Planck(1911)**

Proof of the Clausius Inequality

$$\sum_{i=1}^n \frac{Q_i}{T_i} \leq 0 \quad \text{or} \quad \oint \frac{\delta Q}{T} \leq 0.$$

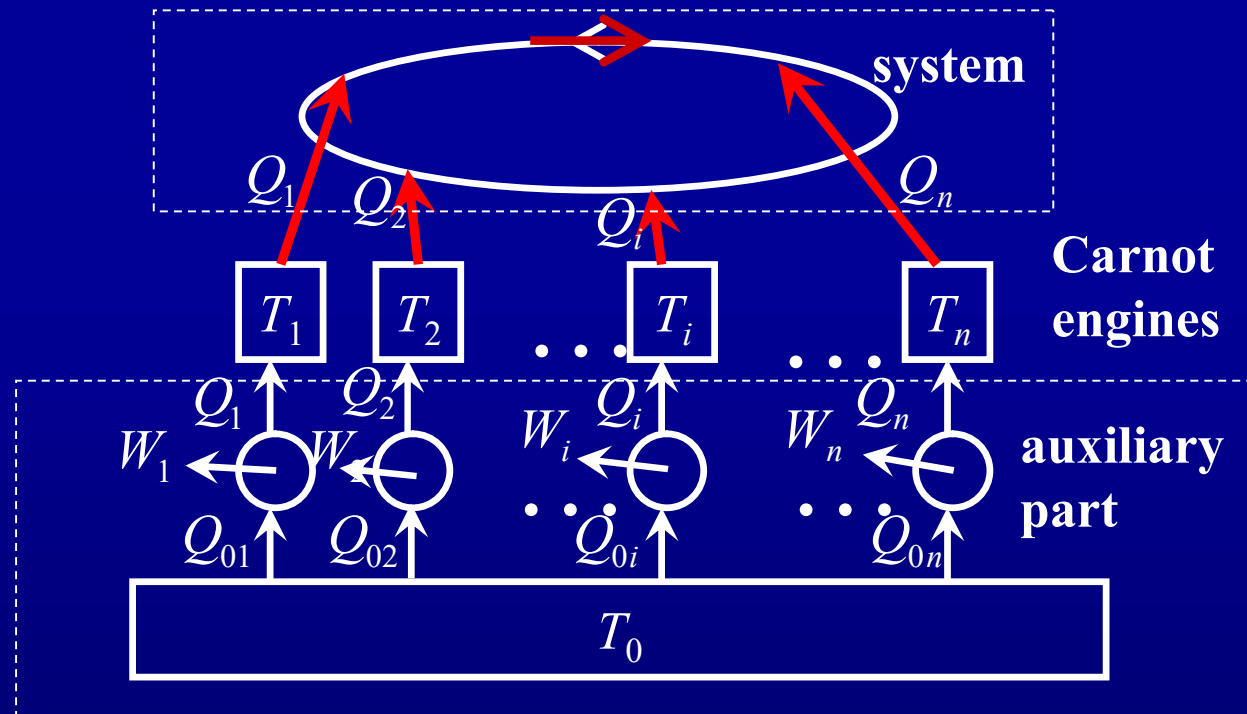




Carnot engines: $\frac{Q_{10}}{T_0} = \frac{Q_1}{T_1}, \dots, \frac{Q_{i0}}{T_0} = \frac{Q_i}{T_i}, \dots, \frac{Q_{n0}}{T_0} = \frac{Q_n}{T_n}.$

First law: $\sum_{i=1}^n Q_{0i} = \sum_{i=1}^n W_i.$

Second law: $\sum_{i=1}^n Q_{0i} = T_0 \sum_{i=1}^n \frac{Q_i}{T_i} = \sum_{i=1}^n W_i \leq 0. \quad \Rightarrow \sum_{i=1}^n \frac{Q_i}{T_i} \leq 0.$



Reversible cycle: $Q_i \rightarrow -Q_i, \quad i = 1, 2, \dots, n.$

$$\sum_{i=1}^n \frac{Q_i}{T_i} \leq 0, \quad \sum_{i=1}^n \frac{-Q_i}{T_i} \leq 0 \Rightarrow \sum_{i=1}^n \frac{Q_i}{T_i} \geq 0.$$

$$\sum_{i=1}^n \frac{Q_i}{T_i} \geq 0, \quad \Rightarrow \sum_{i=1}^n \frac{Q_i}{T_i} = 0 \quad \text{for reversible cycle.}$$